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Arinze Elenasulu

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Python Backend Engineer | AI & Automation

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Professional Summary
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I have been writing Python since 2020 and building backend systems that handle real operational data. I built the backend for a pharmacy operations platform from scratch, designed APIs that process inventory and payment workflows, and sat with end users to turn manual processes into software. I also built CaseFRAME, applying structured AI memory to financial crime investigation workflows where state loss across analyst handoffs is a genuine risk. I write clean, well-structured code and I am looking for a backend engineering role in financial technology where data correctness and auditability matter.

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Core Skills
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Backend Engineering: Python (primary, since 2020), FastAPI, SQLAlchemy, Alembic, REST API design, async patterns, PostgreSQL, schema design, query optimisation

AI & Automation: Agentic workflow orchestration, MCP tooling, LLM integration, RAG pipeline design, structured output design, AI governance and project memory systems

Infrastructure & Reliability: Docker, Linux, structured logging, input validation, defensive backend design, error handling at the system boundary

Testing & Quality: Advanced pytest (fixtures, parametrisation, integration coverage), structured test design, API test automation

Additional Languages & Stacks: Java, Spring Framework, C++, C, JavaScript/TypeScript, MERN stack, MEAN stack, Flutter/Dart (from formal training. Python is my strongest)

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Experience
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Vigilis, Software Engineer

August 2024 – February 2026 · Lagos, Nigeria

Built and maintained the backend for a pharmacy operations platform handling sales, inventory, payments, and reporting across multiple locations. System used day-to-day by pharmacy staff.

- Designed the full data model and API layer (FastAPI, PostgreSQL, SQLAlchemy) with role-based access, JWT auth, and Alembic migrations that kept the database evolving safely as features shipped.
- Sat with pharmacy staff directly to understand their manual processes. Turned paper workflows and mental checklists into software that cut daily admin time and reduced stock discrepancies.
- Built the invoice lifecycle: create, add line items, finalise with automatic stock deduction, cancel with stock reversal. Every state transition validated and audited.
- Added structured logging, input validation, and business rule enforcement that caught errors at the API boundary before they could corrupt data downstream.
- Owned the backend end to end: feature planning, database design, deployment, monitoring, and production bug fixes. No handoff.

Aptech Computer Education, Software Engineer Intern

September 2022 – August 2024 · Lagos, Nigeria

Completed the Advanced Diploma in Software Engineering (ADSE Java track), a structured two-year programme combining classroom training with hands-on project work.

- Covered the full software engineering stack across four semesters: Linux system administration, Python, C++, C, Java, Spring Framework, Enterprise Java Beans (EJB), Flutter/Dart for mobile, and both MERN and MEAN stacks for full-stack web development.
- Built multiple projects across different technology stacks as part of the programme curriculum, gaining practical exposure to how different languages and frameworks solve similar problems.
- Developed strong fundamentals in object-oriented programming, database design, and software architecture that carried into every role since.
- Python became my primary language during this period and has remained so. I have been studying and building with it continuously since 2020.

Bucca Hut, AI and Backend Engineer (Contract, part-time)

February 2025 – May 2025

Built a data analysis tool for a food business, connecting LLM outputs to real operational decisions. Designed the backend API bridging AI processing with the existing workflows.

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Selected Projects

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FRAME, Typed Project Memory Format for AI Agents

Python, Markdown/YAML, Multi-Agent

github.com/haxsysgit/FRAME

Designed a portable standard that gives AI coding agents structured project context across sessions.

- Five-part memory model (Facts, Rules, Acts, Map, Expect) capturing project truth, agent rules, session history, module relationships, and expected outcomes.
- Reduces repeated prompting, context rot, and full-repo rescans across agent sessions.
- Designed as a portable standard: any AI agent reads the same project representation.
- Published docs, learn materials, and frame-python implementation track on GitHub.
- Separation of concerns: FRAME is the contract, implementations earn alignment with it.

Haxaml, AI Agent Governance & Project Memory

Python, MCP, FRAME

github.com/haxsysgit/haxaml · pypi.org/project/haxaml

Built and published a governance runtime that enforces project rules for AI coding agents.

- Governance runtime enforcing FRAME correctness contract, blocks AI-generated code until mechanical checks pass.
- Five-stage pipeline: Admission → Scoped Context → Agent Work → Verification → Recording.

- MCP server so AI coding tools can read and write to FRAME files during a session.
- Local dashboard for inspecting agent session history and project state.
- Published on PyPI with documentation, setup flow, and adoption guides.

Pharmax, Pharmacy Management Platform

Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic

Built the backend for a pharmacy operations platform handling product inventory, sales, invoicing, and reporting.

- Products CRUD with multi-unit pricing (tablets, packs, strips).
- Full invoice lifecycle: draft to finalise (auto stock deduction) to cancel (stock reversal).
- Stock adjustments with audit trail and reorder-level tracking.
- Role-based access with JWT authentication.
- Comprehensive pytest suite covering endpoints, business logic, and edge cases. Sat with pharmacy staff to understand manual processes and turned them into software.

CaseFRAME, AI Continuity for Financial Crime Investigations

Python, FRAME, YAML

github.com/haxsysgit/CaseFRAME

Applied the FRAME memory model to financial crime investigations where state loss across sessions is a genuine operational risk.

- Python loader and validator for structured case packs from YAML.
- Concrete example: reopened sanctions review with unresolved counterparty links.
- State-resumption model , investigation reopens from real case state, not from zero.

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Education

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BSc Information Technology , Middlesex University London (Expected June 2026)

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Advanced Diploma in Software Engineering (ADSE Java) , Aptech Computer Education, Lagos (September 2022 – August 2024)

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